Name: Key

1. a) Compute the following matrix-vector multiplication over the real numbers.

$$\begin{pmatrix} 1 & 8 & 7 & 9 \\ 9 & 2 & 4 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 + 24 - 7 + 0 \\ 9 + 6 - 4 + 0 \\ 0 + 0 + 0 + 0 \end{pmatrix}$$

$$= \begin{pmatrix} 18 \\ 11 \\ 0 \end{pmatrix}$$

b) Give an example of a real matrix that sends u to v, and state the domain and codomain of the corresponding linear transformation; or explain why this is not possible.

$$u = \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 10 \\ 20 \end{pmatrix}$$

For instance,

$$\begin{pmatrix} 2 & 0 & 3 \\ 4 & 0 & -6 \end{pmatrix} \begin{pmatrix} 5 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 \\ 20 \end{pmatrix}$$

Name: Key

2. a) For each of the following matrix operations, calculate the result if the operation is defined. Otherwise, state that the operation is undefined.

$$A = \begin{pmatrix} 4 & 0 & 3 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 2 & 0 & 1 & 1 \end{pmatrix} \quad C = \begin{pmatrix} 2 & 0 & -6 \\ 0 & 1 & 2 \end{pmatrix} \quad D = \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}$$

$AB, \quad A + B, \quad CD, \quad BD$

AB is not defined.

$A+B = \begin{pmatrix} 4 & 0 & 3 & 3 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 4 & 4 \end{pmatrix}$

CD is not defined.

$BD = \begin{pmatrix} 2 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 3 & -2 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \end{pmatrix}$

b) Give an example of two matrices whose product equals M; or explain why this is not possible.

$$M = \begin{pmatrix} 1 & 0 & 3 \\ 4 & 5 & 3 \end{pmatrix}$$

Many answers are possible. For instance,

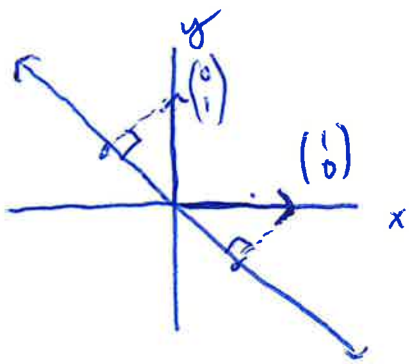
$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 4 & 5 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 3 \\ 4 & 5 & 3 \end{pmatrix}$$

Name: Key

3. a) Find the 2×2 matrix for the following linear transformation using the standard basis for $\mathbb{R}^2$.

project to the line $x + y = 0$

or $y = -x$



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \longmapsto \begin{pmatrix} 1/2 \\ -1/2 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \longmapsto \begin{pmatrix} -1/2 \\ 1/2 \end{pmatrix}$$

So the matrix is:

$$\begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}$$

b) Find the 3×3 matrix for the following linear transformation using the standard basis for $\mathbb{R}^3$.

reflect in the xy-plane, then reflect in the yz-plane

second

first

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

fixes yz-plane, negates x-axis

fixes xy-plane, negates z-axis

Name: Key

Skills Quiz

4. a) Use row reduction to put the following matrix in reduced row echelon form.

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ -1 & -1 & -1 \end{pmatrix} \xrightarrow{\text{row reduction}} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

b) Write the following linear system over $\mathbb{R}$ as an augmented matrix. Then find the general solution to the system and write it in vector form.

$$\begin{aligned} x + 2y - 3z &= -13 \\ x - 4z &= -30 \\ x + 4y - 2z &= 4 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 1 & 2 & -3 & -13 \\ 1 & 0 & -4 & -30 \\ 1 & 4 & -2 & 4 \end{array}\right) \xrightarrow{\text{row reduction}} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left(\begin{array}{ccc|c} 1 & 0 & -4 & -30 \\ 0 & 1 & 1/2 & 17/2 \\ 0 & 0 & 0 & 0 \end{array}\right) \end{array}$$

So
$$\begin{aligned} x \quad - 4z &= -30 \\ y + 1/2\, z &= 17/2 \\ z &= z \end{aligned} \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -30 \\ 17/2 \\ 0 \end{pmatrix} + z \begin{pmatrix} 4 \\ -1/2 \\ 1 \end{pmatrix}$$

Name: Key

5. a) Compute the inverse of each of the following matrices; or explain why an inverse does not exist.

$$A = \begin{pmatrix} 3 & 9 \\ 2 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

A is not invertible, since it has linearly dependent columns.
(Or: the determinant of A is 0.)

For B:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R2-2R1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{\frac{1}{2}R3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1/2 \end{array}\right) \qquad \text{So } B^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}$$

b) Give an example of two matrices A and B such that A is invertible, B is invertible, but AB is not invertible; or explain why this is not possible.

This is not possible. If A has inverse A^{-1} and B has inverse B^{-1}, then AB has inverse $B^{-1}A^{-1}$ since:

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

Name: Key

6. a) Give a basis for the column space of A.

$$A = \begin{pmatrix} 1 & 4 & -5 & 50 \\ 2 & 4 & -4 & 60 \\ 3 & 2 & 0 & 50 \end{pmatrix}$$

One approach is to row reduce A, find the pivot columns, and then take those columns of A. They must span col A and be linearly independent.

$$\begin{pmatrix} 1 & 4 & -5 & 50 \\ 2 & 4 & -4 & 60 \\ 3 & 2 & 0 & 50 \end{pmatrix} \xrightarrow{\text{row reduce}} \begin{pmatrix} \boxed{1} & 0 & 1 & 10 \\ 0 & \boxed{1} & -3/2 & 10 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

so $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 4 \\ 2 \end{pmatrix} \right\}$ is a basis for col A.

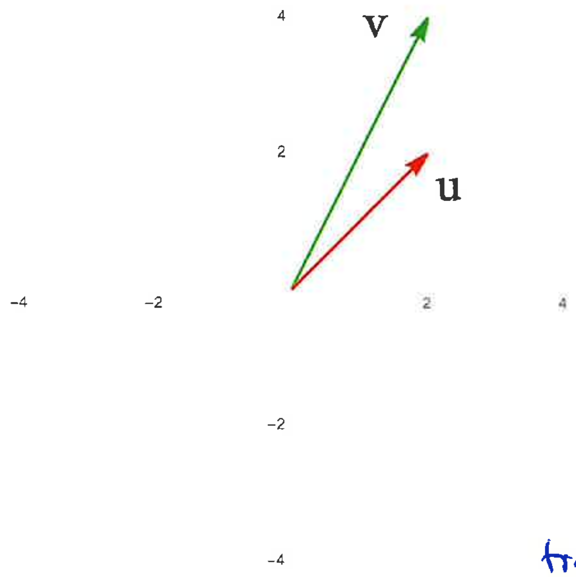
spans $\mathbb{C}^3$ and is linearly independent.

b) Give an example of a basis for $\mathbb{C}^3$ containing the vectors $(i, i, i)^T$ and $(0, 1, -1)^T$ or explain why this is not possible.

Note that $\begin{pmatrix} i \\ i \\ i \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ are linearly independent, since for $k\begin{pmatrix} i \\ i \\ i \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ in the first coordinate, we would have to have $k=0$. But $k\begin{pmatrix} i \\ i \\ i \end{pmatrix} \neq \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ for $k=0$. Since a basis for $\mathbb{C}^3$ must contain 3 vectors, we want one additional vector that is linearly independent from the given pair. Many examples are possible. For instance, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Since $\begin{pmatrix} 1 & i & 0 \\ 0 & i & 1 \\ 0 & i & -1 \end{pmatrix}$ is row equivalent to $\begin{pmatrix} 1 & i & 0 \\ 0 & i & 1 \\ 0 & 0 & -2 \end{pmatrix}$, which has 3 pivots, the set $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} i \\ i \\ i \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$

Name: Key.

7. a) Using the standard basis, give the matrix for the linear transformation of $\mathbb{R}^2$ that sends $\mathbf{u}$ to $\mathbf{v}$ and sends $\mathbf{v}$ to $2\mathbf{u}$. The vectors are pictured below, in the standard coordinate system.



In the basis u, v, the matrix representing this linear transformation is

$$\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$$

To write the same linear transformation in the standard basis, we should:

① convert from the standard basis to the u, v basis

② carry out the linear transformation in the u, v basis

③ convert back to the standard basis from the u, v basis.

Since the matrix that converts from the u, v basis to the standard basis is $\begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$ — it takes u "$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$" to $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$ (its name in the standard basis) and v "$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$" to $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$, we have that the desired matrix in the standard basis is:

$$\begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}^{-1} = \boxed{\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}}$$

Check: $\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}\overset{u}{\begin{pmatrix} 2 \\ 2 \end{pmatrix}} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = v.$ ✓

$\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}\overset{v}{\begin{pmatrix} 2 \\ 4 \end{pmatrix}} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 2u.$ ✓

b) Give an example of a matrix that is similar to an elementary matrix but that is not itself an elementary matrix; or explain why this is not possible.

For instance, $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ is an elementary matrix,

and $$\begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 11 & -20 \\ 4 & -7 \end{pmatrix}$$

is similar to $\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$ (by definition) but is not an elementary matrix.

Name: Key

8. a) Compute the determinant of this matrix:

$$A = \begin{pmatrix} 1 & 1 & 0 & 6 \\ -1 & 2 & 0 & 5 \\ 0 & -1 & 2 & 0 \\ 3 & 1 & 0 & 1 \end{pmatrix}$$

For instance,
by cofactor expansion we have:

$$\det(A) = +2 \cdot \begin{vmatrix} 1 & 1 & 6 \\ -1 & 2 & 5 \\ 3 & 1 & 1 \end{vmatrix} = 2\left(1 \cdot \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} - 1 \cdot \begin{vmatrix} -1 & 5 \\ 3 & 1 \end{vmatrix} + 6 \cdot \begin{vmatrix} -1 & 2 \\ 3 & 1 \end{vmatrix}\right)$$

$$= 2\left(-3 - (-16) + (-42)\right) = 2(-29) = \boxed{-58}$$

b) Give two examples A and B of 2×2 matrices with determinant -1. Then using your examples, calculate $\det(AB)$ and $\det(A^{25})$.

$A = \begin{pmatrix} 5 & 3 \\ 2 & 1 \end{pmatrix}$, $\det A = (5 \cdot 1) - (2 \cdot 3) = 5 - 6 = -1$

$B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$, $\det B = (-1)(1) - (0)(0) = -1$

$\det(AB) = (\det A)(\det B) = (-1)(-1) = 1$, so $\boxed{\det AB = 1}$

$\det(A^{25}) = (\det A)^{25} = (-1)^{25} = -1$, so $\boxed{\det(A^{25}) = -1}$

Name: Key

9. a) Find the eigenvalues and eigenvectors of this matrix:

$$A = \begin{pmatrix} 10 & 3 \\ 1 & 12 \end{pmatrix}$$

$$\det(A - \lambda I) = \det\begin{pmatrix} 10-\lambda & 3 \\ 1 & 12-\lambda \end{pmatrix} = (10-\lambda)(12-\lambda) - 3$$

$$= \lambda^2 - 22\lambda + 117$$

If $\lambda^2 - 22\lambda + 117 = 0$, we have $(\lambda - 13)(\lambda - 9) = 0$

$$\Rightarrow \lambda = 13 \text{ or } 9.$$

$\lambda = 13$: $\begin{pmatrix} -3 & 3 \\ 1 & -1 \end{pmatrix} \to \begin{pmatrix} -3 & 3 \\ 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$, so the 13-eigenspace is $\text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$

$\lambda = 9$: $\begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix} \to \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$, so the 9-eigenspace is $\text{span}\left\{\begin{pmatrix} -3 \\ 1 \end{pmatrix}\right\}$

$$\left\{\begin{array}{l} x = -3y \\ y = y \end{array}\right. \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = y\begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

b) Give an example of two matrices A and B with the same set of eigenvalues, but where A is diagonalizable but B is not; or explain why this is not possible.

For instance,

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{pmatrix}$$

Other examples

$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

Both have eigenvalues 2 and 5.
For A, for each eigenspace we have alg mult = geom mult.
But for B, the 5-eigenspace has alg mult 2 but geom mult 1.